

SCHUR TENSOR PRESERVATION ANSWERS A DUNFORD–PETTIS TENSOR QUESTION

RUN FA_BANACH_001

Status. `literature_already_answered`. This is not a new proof. It records that a later theorem of Rodriguez answers the final tensor square questions in Gonzalez–Gutierrez arXiv:math/0004101.

SOURCE QUESTION

Gonzalez–Gutierrez, *The Dunford–Pettis property on tensor products*, arXiv:math/0004101, consider an infinite-dimensional separable $\mathcal{L}_\infty$ -space X with the Schur property such that X^* is isomorphic to $C[0, 1]^*$. In the final ‘Questions’ environment, labelled `qq` in the source, they ask whether

$$X \widehat{\otimes}_\pi X$$

has the Schur property, and whether it has the Dunford–Pettis property.

SUPPORTING ANSWER

Rodriguez, *On weak compactness in projective tensor products*, arXiv:2206.08651, explicitly cites the Schur-preservation problem for projective tensor products as open, citing Gonzalez–Gutierrez among the sources. Immediately afterward, Theorem `theo:Schur` states:

if X and Y have the Schur property, then so does $X \widehat{\otimes}_\pi Y$.

Applying this theorem with $Y = X$ gives that the source paper’s Bourgain–Delbaen space satisfies that $X \widehat{\otimes}_\pi X$ has the Schur property. Since every Banach space with the Schur property has the Dunford–Pettis property, $X \widehat{\otimes}_\pi X$ also has the Dunford–Pettis property.

SCOPE

This answers the two final questions in arXiv:math/0004101 for the specified Schur $\mathcal{L}_\infty$ -space X : both answers are affirmative. The packet does not claim a new proof and does not address other unrelated Dunford–Pettis tensor-product problems from the source paper.

SEARCH AND PROVENANCE NOTES

The run indexes were searched for arXiv:math/0004101 and phrases around the Dunford–Pettis property, projective tensor products, injective tensor products, and Schur tensor preservation. No existing packet for the exact source question was found. A relevant conditional packet for arXiv:1504.00520 uses a hypothetical negative answer to Schur tensor preservation as a dependency; Rodriguez’s theorem gives the positive general preservation result, so that negative route is no longer viable.

REFERENCES

- [1] M. Gonzalez and J. M. Gutierrez, *The Dunford-Pettis property on tensor products*, arXiv:math/0004101.
- [2] J. Rodriguez, *On weak compactness in projective tensor products*, arXiv:2206.08651.